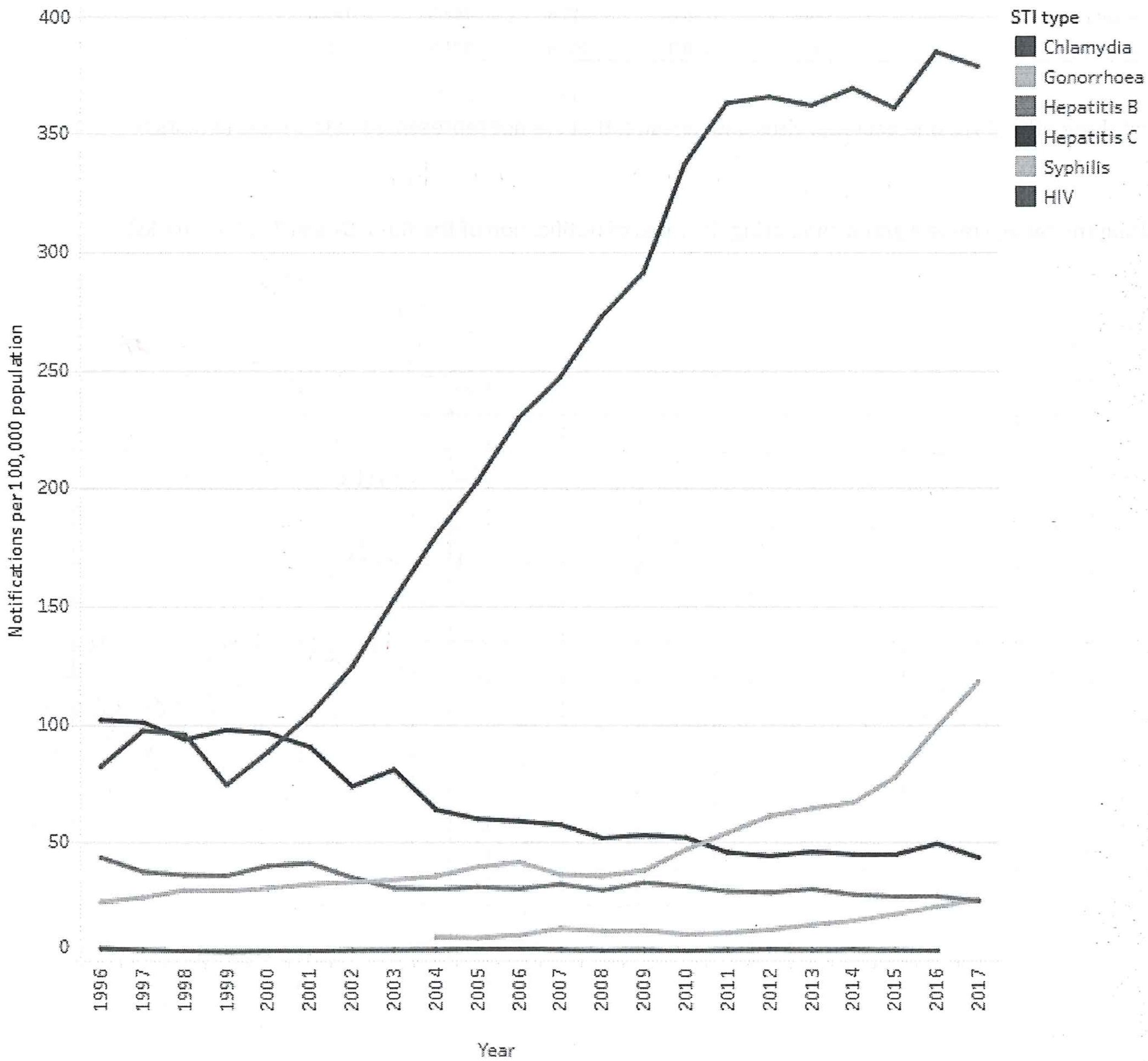
Marking Key

STIs assessment – Science Inquiry Skills **Name** ____________________

10% of yearly grade

Notification rates of sexually transmissible infections and blood-borne viruses, 1996–2017



Using data from the graph, what was the percentage increase in Chlamydia infections from 1999 to 2017? Show your working (3 marks)

2017 ~ 375
1999 ~ 75 (1)

375 − 75
= 300. (1)

$\frac{300}{75} \times 100 = 400\%$

Using the graph, calculate the increase in rate per year from 2003-2009 (hint – rise over run) (2 marks)

rise = (~150 to ~275) ~125 or 150 (1)

run = 6 yrs

$\frac{125 \text{ or } 150}{6} = 20.8$ /yr. increase

Notification rates of selected sexually transmitted diseases, Australia, 2013–2017

Notifiable disease	2013	2014	2015	2016	2017
Gonorrhoea	64.4	66.8	77.6	98.7	93.7
Genital Herpes	124.6	128.8	143.6	142.3	147.0
Chlamydia	362.0	369.3	352.5	371.5	312.0

pubic lice/scabies

List three other STIs, one bacteria, virus and parasite, that are not represented in this table (3 marks)

Syphilis HIV/AIDS, HPV, Hepatitis, etc

Using the table, create a graph comparing the rates of notification of the three STIs in 2017. (5 marks)

1 type - Bar

2 labels inc. units

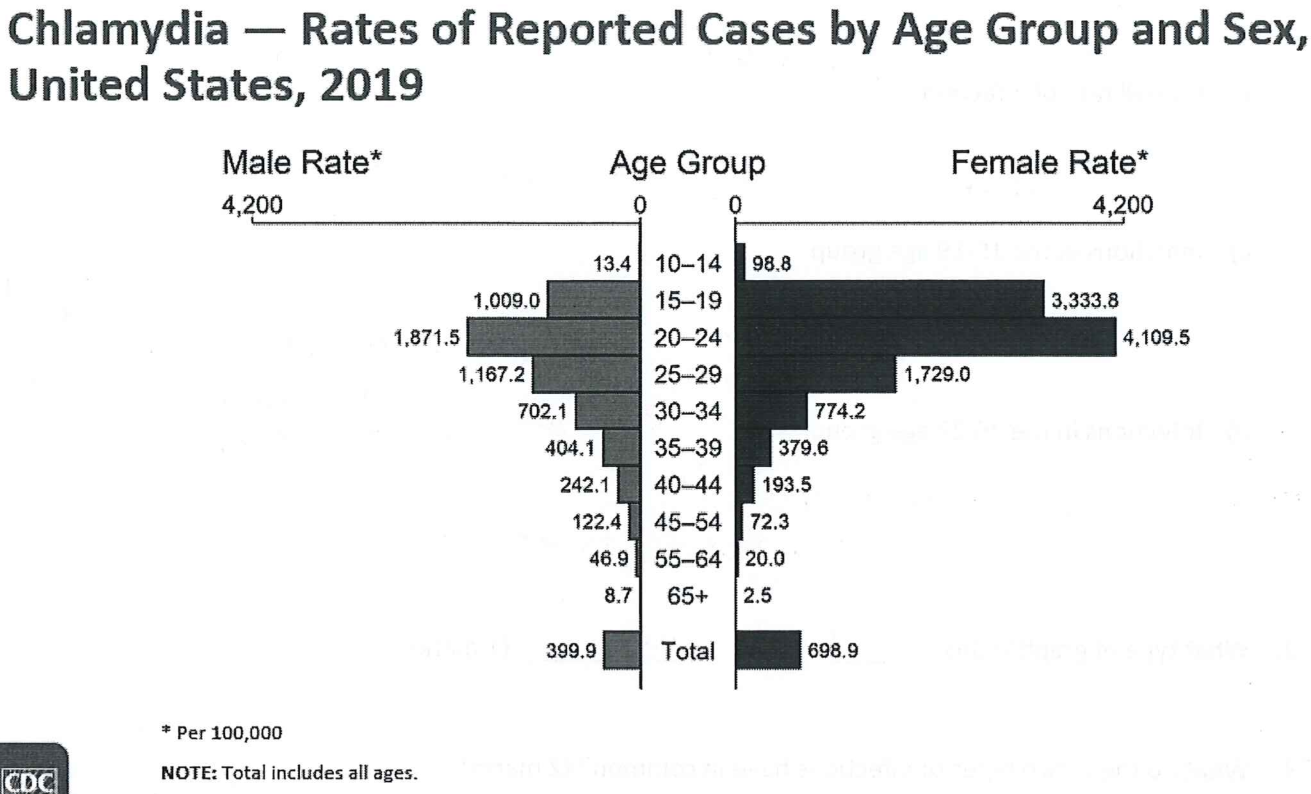
3 scale

4 title

5 correct data - only 2017 & correct plot.

The questions on the following page refer to the graph and table on this page.

Below is a graph outlining the rate of Chlamydia infections in the US.



Below is a table outlining the same set of data but for Gonorrhoea.

Year	Age Group	Sex	Rate
2019	10–14	Female	20.7
2019	15–19	Female	559.5
2019	20–24	Female	737.4
2019	25–29	Female	444.1
2019	30–34	Female	265.4
2019	35–39	Female	154.2
2019	40–44	Female	84.9
2019	45–54	Female	32.6
2019	55–64	Female	8.8
2019	65+	Female	1.2
2019	Total	Female	152.6

Year	Age Group	Sex	Rate
2019	10–14	Male	4.9
2019	15–19	Male	328.6
2019	20–24	Male	743.5
2019	25–29	Male	700.6
2019	30–34	Male	524.9
2019	35–39	Male	341.7
2019	40–44	Male	223.2
2019	45–54	Male	128.3
2019	55–64	Male	57.3
2019	65+	Male	10.1
2019	Total	Male	224.4

1. Using data from the previous page, outline the differences between chlamydia and gonorrhoea on the basis of: (8 marks)
   a) Sex

   Chlamydia : ↑ females
   G : ↓ females.

   b) Overall rate of infection

   Chlamydia higher than Gonorrhoea.

   c) Infections in the 15-19 age group

   higher in chlamydia (1).
   2nd highest overall in chlamydia but only 3rd highest in Gonorrhoea.

   no data: -1.

   d) Infections in the 25-29 age group

   higher in chlamydia (1)
   opposite to above.

2. What type of graph is this? histogram (1 mark)

3. What do these two types of infections have in common? (2 marks)

   bacterial treatment
   transmission

4. Although the rates of chlamydia may be higher, why may gonorrhoea infections be more of a concern? (1 mark)

   antibiotic resistance/ difficult to treat

5. Give two possible reasons, other than higher rates of infection, that may have caused higher rates of detection/reporting of Chlamydia in females rather than males. (2 marks)

   - more noticeable / less 'silent'
   - more likely to get tested
   ~~more~~

6. The rate is based on number of cases per 100,000. How many males aged 15-19 would be expected to be diagnosed with gonorrhoea in a city with 20,000 people? Show your working (3 marks)

   $328.6 \times \frac{1}{5} = 65.72$